

String Interpolation with Template Strings

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Contents

- [1 Introduction](#)
 - [1.1 Prior Work](#)
 - [1.2 Prior Art in Other Languages](#)
- [2 Design](#)
 - [2.1 Lexing](#)
 - [2.2 Handling Macro Expansion](#)
 - [2.3 Lexing Trailing Equals](#)
 - [2.4 Lexing Nested Expressions](#)
 - [2.5 Concatenating Consecutive String Literals](#)
 - [2.6 Data Members](#)
 - [2.7 Interpolation Information](#)
 - [2.8 Examples](#)
 - [2.9 The TemplateString Concept](#)
 - [2.10 Implementation Experience](#)
 - [2.11 More Formal Lexing Specification](#)
- [3 Alternate Approaches](#)
 - [3.1 f-strings](#)
 - [3.2 Reduced Representation](#)
 - [3.3 Redundant Information](#)
 - [3.4 Static Data Members or Static Member Functions](#)
 - [3.5 Wait for Reflection](#)
- [4 References](#)

§ 1 Introduction

The `std::format` approach to formatting offers many significant benefits over the prior `<iostream>`s approach that need not be revisited here. However, `<iostream>` does still have one significant advantage: ordering. It is easy to see at a glance with a long `std::cout` statement which pieces are to be formatted in which order. With `std::format`, as the amount of replacement fields increases, it becomes increasingly difficult to ensure that they are all correctly ordered.

The solution to this problem is string interpolation: the ability to put the expression to be formatted inside of the format string. This lets us preserve all of the advantages of `std::format`, while also regaining ordering. String interpolation is a wildly popular language feature due to the ease with which it allows users to express complex ideas. It's not surprising that a huge number of modern languages support this to some degree or another. A non-exhaustive list includes: C#, D, Elixir, F#, Groovy, Kotlin, JavaScript, Perl, PHP, Python, Ruby, Rust, Scala, Swift, and VB.

§ 1.1 Prior Work

There have been two prior WG21 papers pursuing string interpolation as a C++ language feature: [\[P1819R0\] \(Interpolated Literals\)](#) and [\[P3412R3\] \(String Interpolation\)](#). The two proposals are quite different, so let's consider an example to work through the details:

```
auto get_result() -> int { return 42; }

auto example() -> void {
    auto interp = f"The result is {get_result()}\n";    // #1
    std::print(interp);                                // #2
    std::print(f"The result is {get_result()}\n");      // #3
}
```

In P1819, the `interp` is an object that is roughly equivalent to:

```
auto interp = [&](auto&& f) -> decltype(auto) {
    return f("The result is ", get_result(), "\n");
};
```

So line #1 does nothing. The call to `get_result()` does not happen yet. Instead, the library would provide new overloads of `std::print` and friends so that in line #2, the library would invoke `interp` with the appropriate function to do the printing. The call to `get_result()` happens at that point. Line #3 does the same things as lines #1 and #2, just together.

In P3412, the behavior is very different. `interp` is already a `std::string`, which is evaluated as:

```
auto interp = std::format("The result is {}\n", get_result());
```

This makes the call in line #2 ill-formed, since `std::print` cannot accept a `std::string`. However, the call in line #3 is valid — by way of a change to overload resolution that recognizes this case as special and instead evaluates the call directly as:

```
std::print("The result is {}\n", get_result());
```

In short, P1819 gives us an object (that doesn't evaluate any of the expressions) while P3412 gives us either a `std::string` or an argument list, depending on context.

Of the two, I think P1819 is significantly better. We get a simple object that can allow for a wide variety of potential functionality. It has two big problems though. The first is that it evaluates lazily and stores its data opaquely — which leads to more surprising behavior, the potential for dangling references, and arbitrarily limited usage. The other is its breakup into pieces doesn't actually play very well with `std::format` — where we would want there to be a format string and we don't have one. The original motivation for the lambda approach was ease of use — get all the expressions in one convenient format. But the language has evolved since 2019. We have both reflection and packs in structured bindings now, so we don't need the lambda approach anymore.

On the other hand, P3412 is a very complex design, with an overload resolution mechanism that is very strongly coupled to the current implementation strategy of formatting. What if we someday get `constexpr` function parameters and it turns out to be better to implement `basic_format_string<char, Args...>` as taking a `constexpr` `string_view` instead of it being a `constexpr` constructor? Would we need to change how string interpolation is defined too? All this complexity buys us is the ability to create a `std::string` in a single character. However, I don't think that's even a good goal for C++ — we shouldn't hide an operation as costly as string formatting in a single character, and we shouldn't tie the language so tightly to this particular library (as P3412 does via `__format__` directly calling `std::format`).

Instead, this paper proposes an idea much closer to the P1819 model.

§ 1.2 Prior Art in Other Languages

Python 3.6 introduced literal string interpolation (`f"..."`) in [PEP-498], which was later extended in Python 3.14 by template strings (`t"..."`) in [PEP-750]. The former directly produces a `string`, while the latter gives a template string — an object with enough information in it to be formatted later. This is similar to Rust's `format_args!`, which gives you a completely opaque object (unlike Python's which is completely specified). Both languages gives you a facility to take an interpolated string and produce an object for future work (similar to P1819).

JavaScript also has [template literals](#), which support tagging. A tagged template literal is quite similar to what [P3412R3] proposes: the expression `myTag`That ${person} is a ${age}.` evaluates as `myTag(["That ", " is a ", "."], person, age)`, similar to the C++ proposal having `myTag(f"That {person} is a {age}")` evaluate the transformed call `myTag("That {} is a {}", person, age)`. Here, the literal is *not* an object.

§ 2 Design

This paper proposes that we introduce string interpolation for C++ following the same idea as Python's template strings. We'll have to come up with a better name than “template string” for this, but for now I'm going to stick with it. A template string literal will eagerly evaluate all of the expressions and produce a new object with all of the relevant pieces, such that it will be suitable for both formatting APIs (like `std::format` and `std::print`) and other APIs that have nothing to do with formatting.

Our example from earlier:

```
auto get_result() -> int { return 42; }

auto example() -> void {
    auto interp = t"The result is {get_result()}\n";
}
```

will evaluate as:

```
auto example() -> void {
    struct Template {
        static consteval auto fmt() -> char const* { return "The result
is {}\\n"; }
        static consteval auto num_interpolations() -> size_t { return 1;
        }
        static consteval auto string(size_t i) -> char const* {
            constexpr char const* data[] = {"The result is ", "\\n"};
            return data[i];
        }
        static consteval auto interpolation(size_t i) -> std::interpolat
        ion {
            constexpr std::interpolation data[] = {"get_result()", "
", 0, 1}};
            return data[i];
        }

        int _0;
    };

    auto interp = Template{get_result()};
}
```

Let's go through all of these pieces in order. A template string will generate an instance of a not-necessarily-unique type (unlike a lambda expression, which always has a unique type) by parsing the replacement fields of the string literal. The parsing logic here is very basic and does not need to understand very much about either the format specifier mini-language or C++ expressions more broadly. The interpolation type will have five public pieces of information in it:

- `fmt()` is a static member function that returns the format string,
- `num_interpolations()` is a static member function that returns the number of interpolations (possibly 0),
- `string(i)` is a static member function that returns the *i*th string part
- `interpolation(i)` is a static member function that returns the *i*th [interpolation information](#), and then lastly
- one non-static [data member](#) for each (possibly-nested) expression. The name of the member is unspecified, but the order and types are.

With this structure, we can easily add additional overloads to the format library to handle template strings:

```
template <TemplateString S>
auto print(S&& s) -> void {
    auto& [...exprs] = s;
    std::print(s.fmt(), exprs...);
}
```

Note that there is no difference in handling between lines #1–2 and line #3 in the example (like the P1819 design and unlike the P3412 one). A template string is just an object, that contains within it all the relevant information. So whether we construct the object separately doesn't matter.

The rest of the paper will go through details on first on how the parsing works and then into other examples to help motivate the structure.

Keep in mind that since a template string object is *just an object*, where most of the information are static data members, this ends up being a very embedded-friendly design too.

§ 2.1 Lexing

A template string is conceptually an alternating sequence of string literals and interpolations. The string literal parts are just normal string literals — we look ahead until we find a non-escaped `{` to start the next replacement field ("`{{`" in a format string is used to print the single character `'{'`). A replacement-field comes in two forms:

```
replacement-field:
    { expr }
    { expr : format-spec }
```

For example, a template string like `t"The price of {id:x} is {price}."` needs to be lexed into these five pieces:

String Literal	"The price of "
Interpolation	"{:x}" with expression id
String Literal	" is "
Interpolation	"{}" with expression price
String Literal	."

The first and last piece are always (possibly-empty) string literals — for N interpolations there will be $N+1$ strings. Even for the template string `t"{expr}"` which entirely consists of an expression, there will be two empty string pieces.

However, C++ expressions can be arbitrary complicated. Notably, they can also include `:` or `}`, both of which are significant in formatting and indicate the end of the expression. So how do we know when

we're done with the *expr* part here?

One approach would be to simply limit the kinds of expressions that can appear in template strings. Rust, for instance, *only* supports identifiers. That obviously makes parsing quite easy, but it also is very limiting. On the other extreme, supporting *all* expressions can easily lead to indecipherable code. I think on balance, supporting only identifiers is far too restrictive. But once you start adding what other kinds of expressions to allow (surely, at least class member access), it quickly becomes too difficult to keep track of what is allowed (indexing? function calls? splices?) and ironically makes both the implementation more difficult (to enforce what is and isn't allowed) and harder to understand for the user (to know which expressions are and aren't allowed).

I think it's best to simply allow anything in the expression (as Python does) and trust the user to refactor their expressions to be as legible as they desire. This allows us to take a very simple approach: we simply lex *expr* as a balanced token sequence (just counting `{}`s, `()`s, and `[]`s), so that colons and braces inside of any of the bracket kinds are treated as part of an expression. But the first `:` or `}` encountered when we're at a brace depth of zero means we're done with *expr*.

Here are some examples of template string formatting calls and how they would be evaluated. The first column will show a template string that consists entirely of an expression. The second column will show how the format string for that expression will be lexed and the third column will show the expression.

template string	lexed format string	lexed expression
<code>t"{x}"</code>	<code>"{}"</code>	<code>x</code>
<code>t"[]{ return 42; }()"</code>	<code>"{}"</code>	<code>[] { return 42; }()</code>
<code>t"{co_await f(x) + g(y) / z}"</code>	<code>"{}"</code>	<code>co_await f(x) + g(y) / z</code>
<code>t"{a::b}"</code>	<code>"{:b}"</code>	<code>a</code>
<code>t"{(a::b)}"</code>	<code>"{}"</code>	<code>a::b</code>
<code>t"{cond ? a : b}"</code>	<code>"{:b}"</code>	<code>cond ? a</code> ❌
<code>t"{(cond ? a : b)}"</code>	<code>"{}"</code>	<code>(cond ? a : b)</code>

Two important things to note here. First, it is possible to lex an invalid expression due to finding a `:` first, as in the penultimate line. The lexing won't know about this though.

Second, note that `"{a::b}"` lexes as simply the expression `a`, not `a::b`. If the latter is desired, it has to be parenthesized. That is, while lexing, we are not simply looking for the *token* `:`, but the *character* `:`. Which could be the token `:` but also includes not just two-character tokens like `::` and `:]`, and even the digraph `:<` (e.g. `"{a:<5}"` lexes as the expression `a` with the format specifier `<5` — left-aligned with width `5` — not as the incomplete expression `a[5]`). This is a difference in the logic proposed in [\[P3412R3\]](#), which looks specifically for the *token* (not character) `:`. This is important because it allows for the most functionality:

```
namespace v { int x = 2; }

auto example() -> void {
    std::vector<int> v = {10, 20, 30};
    std::println(t"{v::x}");    // [a, 14, 1e]
```

```
std::println(t"{{v::x}}"); // 2
}
```

In order to format expressions that use scoping or the conditional operator, you'll just have to write parentheses. That seems easy enough to both understand and use. Otherwise, anything goes.

§ 2.2 Handling Macro Expansion

While lexing expressions, macro expansion occurs too (although at a later phase, [see below](#)). This is both what users expect and is important to support, otherwise we're not actually meeting the claim of supporting all expressions. There are even standard utilities that are defined as macros, like `errno`. The one thing to note is that looking for the terminating `:` or `}` of an expression will *not* consider such a character from macros. Otherwise, the macros wouldn't actually be usable properly and also the format string itself would become illegible.

Extending the previous example:

```
namespace v { int x = 2; }

auto example() -> void {
    std::vector<int> v = {10, 20, 30};
    std::println(t"v::x"); // [a, 14, 1e]
    std::println(t"{{v::x}}"); // 2

    #define SCOPED v::x
    std::println(t"{{SCOPED}}"); // 2, not [a, 14, 1e]
}
```

§ 2.3 Lexing Trailing Equals

One nice debugging feature that Python's f-strings (and template strings) have is the equals suffix:

```
>>> f"{x=}, {y=}, {z=}"
"x=5, y=7, z='hello world'"
```

Concretely, an expression that ends with an `=` (surrounded by any amount of whitespace) has that suffix appended to the previous string literal piece. Occasionally, there are requests to support `std::print(x, y, z)` to just concatenate those three elements — but that's not as useful as it initially seems since you quickly forget what variables you're printing in what order. The ability to support this on the other hand is *very* useful for debugging:

```
std::println(t"{x=}, {y=}, {z=?}");
```

It is also quite easy to implement, since it's just a matter of checking if the last lexed token of the expression was an `=`. And, if so, dropping that from the expression (since no valid C++ expression ends with `=`) and instead adding the stringified expression to the previous string part.

Concretely: `t"Hello {name=}"` behaves exactly equivalently to `t"Hello name={name}"`.

Note that this isn't *quite* what Python does — as you might notice from the Python example. In Python, it behaves like `t"Hello name={name!r}"`, which is basically calling `repr(name)` instead of `str(name)`. It would be really nice if we actually had a real answer for debug formatting — but neither `std::format` nor `fmt::format` really have one. There is a `?` specifier which is used to help ensure that range formatting properly works ([P2286R8]), but it's not valid across all types, and there's no special handling for it. So we can't really make `t"{name=}"` evaluate as `t"name={name:?}"`, since that would only work for a small set of types. Instead, this paper proposes not to add any format specifier here.

§ 2.4 Lexing Nested Expressions

Consider the template string:

```
t"{name:>{width}}"
```

There are two expressions here: `name` and `width`. Or rather, we know `name` is an expression, but how do we know that `width` is? In the format model, the format specifiers can be *anything*. There really are no rules — as long as the type's formatter can handle it. Having nested braces in a format specifier commonly refers to another argument, but it need not actually mean that.

In my CppCon 2022 talk [The Surprising Complexity of Formatting Ranges](#), I work through an example of how one might add underlying specifiers to `std::pair` and `std::tuple`, [where](#):

```
int main() {
    auto elems = std::tuple(10, 20, 30);
    fmt::print("{}\n", elems);                // (10, 20, 30)
    fmt::print(":{x}{#x}{-^4}}\n", elems);    // (a, 0x14, -30-)
    fmt::print(":{x}{}{x}}\n", elems);        // (a, 20, 1e)
}
```

There, `{x}` doesn't refer to the expression `x`, it was just chosen as notational convenience. So how can we get this to work?

```
fmt::print(t"{elems:{x}{}{x}}\n");
```

The short answer is: we cannot. We need to make sense of the template string literal separate from type information, and even if we had type information, it's not like we have a way for the formatter API to signal when it's expecting an expression. We just have to make a choice up front for what to do here. I think we have three options:

1. We could explicitly opt *in* to a nested expression. That is, maybe something like `t"{elems:{x}}${x}"` signals that the first `{x}` is just a string but the second `{x}` is actually the expression `x`. So in the above example, we simply wouldn't use the `$`.
2. We could explicitly opt *out* of being a nested expression with added escaping. Which in this case would be `t"{elems:{{x}}{{}}}{x}}"`.
3. We could say that in a format specifier, encountering a `{` *always* begins an expression that ends at `:` or `}` (same as top-level) and that there is neither opt-in nor opt-out. Meaning that the approach to format specifiers that I demonstrated in that talk wouldn't work, and would instead have to be `t"{elems:{:x}{}{:x}}"`.

I think the third option here is the best. It is at most a minor burden on users as I doubt this approach is in widespread use, and allows for a design that is as simple as possible.

Getting back to our original example:

```
t"{name:>{width}}"
```

This would lex as:

String Literal	""
Interpolation	"{:>{}]" with two expressions: name and width
String Literal	""

And [this works](#) because we recognize `{:x}` as not being a nested expression:

```
fmt::print(t"as template: {elems{:x}{}{:x}}\n"); // as template: (a,
20, 1e)
```

If we simply stored the expressions and the format string, that would be straightforward: we just have two [data members](#). But we can do better than that. But before we get into the [interpolation information](#), I'll talk about the data members.

§ 2.5 Concatenating Consecutive String Literals

Consecutive string literals are concatenated during preprocessing. The same should hold true for template string literals — which can be concatenated with each other and also with regular string literals, in any order. In the table below, imagine we are initializing a variable `s` to the token sequence shown in the first column and examining the resulting `s.fmt()` and the expressions being lexed:

Tokens	<code>s.fmt()</code>	expressions
"Hello, " "World"	n/a	n/a
"Hello, " t"{name}"	"Hello, {}"	1: name
t"{greeting}, " t"{name}"	"{}, {}"	2: greeting, name

Tokens	s.fmt()	expressions
t"{greeting}, " "World"	"{}", World"	1: greeting
t"{greeting}, " "{}"	"{}", {}"	1: greeting ❌

Note the last line. We're concatenating a template string literal and a regular string literal — that simply concatenates the contents of the 2nd string literal onto the last string piece of the 1st — there is no implicit escaping of the braces, so the resulting format string would be incomplete — it has 2 replacement fields but only one expression.

We could consider implicitly escaping the braces for regular string literals that are concatenated to template string literals. I don't know if that's a good idea though.

§ 2.6 Data Members

For any expression, E (including nested expressions — so there may be more expressions than interpolations), a non-static data member will be generated (and then initialized from E) having type `decltype((E))`. This ensures that we get the right type, but also that we're not copying anything unnecessarily. For instance:

```
auto verb() -> std::string;

auto example(std::string const& name, std::string relation) -> void {
    auto tmp1 = t"Hello, my name is {name:?}. You killed my {relation}.
                Prepare to {verb()}.";
}
```

The object `tmp1` will have three members:

- the first, corresponding to `name`, has type `std::string const&`
- the second, corresponding to `relation`, has type `std::string&`
- the third, corresponding to `verb()`, has type `std::string`

These are all public, non-static data members. If we want to copy all of the members (e.g. because we want to serialize them), we can easily do so. It's just that there is no need for the template string itself to do so directly. Note that the call to `verb()` happens immediately and the result is stored in `tmp1`. We are not lazily holding onto the expression `verb()`.

This *does* open up the opportunity for dangling if you write something like this:

```
auto oops(int value) { return t"{value}"; }
```

That template string object will have an `int&` member, refer to the parameter that will be destroyed when we return from the function. I don't think this is a huge use-case of template strings, but it's something to keep in mind. The `decltype((E))` logic is essential for ensuring no overhead for template strings in the

expected use-cases. We could consider something like a *leading =* to capture by value instead of by reference, but users can already write `"t{auto(value)}"` there too.

§ 2.7 Interpolation Information

A template string will have a `static consteval` member function which returns objects of type `std::interpolation`, which is a simple aggregate with four members:

```
struct interpolation {  
    char const* expression;  
    char const* fmt;  
    size_t index;  
    size_t count;  
};
```

`expression` is a string literal that is the stringified version of the expression (before macro expansion). `fmt` is the full format specifier with the expression removed, that you would need in order to format this specific expression. Then, we need both `index` and `count` in order to support nested replacement expressions, as we just went through. That gives us both the first non-static data member and the amount of non-static data members this interpolation is associated with.

The template string literal `t"{name=:>{width}}"` would generate the object

```
struct Template {  
    static consteval auto fmt() -> char const* { return "name={:>{}}"; }  
    static consteval auto num_interpolations() -> size_t { return 1; }  
    static consteval auto string(size_t n) -> char const* {  
        constexpr char const* data[] = {"name=", ""};  
        return data[n];  
    }  
    static consteval auto interpolation(size_t n) -> std::interpolation  
    {  
        consteval std::interpolation data[] = {{  
            .expression = "name", // note that "name" is preserved  
            .fmt = "={:>{}}", // but "width" is not  
            .index = 0,  
            .count = 2  
        }};  
        return data[n];  
    }  
  
    std::string const& _0;  
    int const& _1;  
};
```

Why do we go through the trouble of proving `string(n)` and `interpolation(n)`? Let's look at some examples...

§ 2.8 Examples

Consider again our example from earlier:

```
auto tpl = t"Hello, my name is {name}. You killed my {relation}.  
Prepare to {verb():*^{width}}.";
```

I already showed how we could print this normally, which didn't use either of those two arrays:

```
template <TemplateString S>  
auto print(S&& s) -> void {  
    auto& [...exprs] = s;  
    std::print(s.fmt(), exprs...);  
}
```

But we could do a lot more with this object other than just print it. We could write a function to automatically highlight the interpolations green and bold, which is supported by the `fmt` library but not the standard. That's straightforward, since we're not tied into any particular library, and we have all the information we need:

```
template <TemplateString S>  
auto highlight_print(S&& s) -> void {  
    constexpr size_t N = s.num_interpolations();  
  
    auto& [...exprs] = s;  
  
    template for (constexpr int I : std::views::indices(N)) {  
        fmt::print(s.string(I));  
  
        constexpr auto interp = s.interpolation(I);  
        constexpr auto [...J] = std::make_index_sequence<interp.count>();  
        fmt::print(fmt::emphasis::bold | fg(fmt::color::green),  
                    interp.fmt,  
                    exprs...[interp.index + J]...);  
    }  
  
    fmt::print(s.string(N));  
}
```

Or we could turn it into the JSON object `{"name": "Inigo Montoya", "relation": "father", "verb()": "die"}`:

```
template <TemplateString S>  
auto into_json(S&& s) -> boost::json::object {  
    auto& [...exprs] = s;
```

```

boost::json::object o;
template for (constexpr int I : std::views::indices(N)) {
    constexpr auto interp = s.interpolation(I);
    o[interp.expression] = exprs...[interp.index];
}
return o;
}

```

Or we could use entirely different formatting mechanisms. We could make this work with `printf` too! This isn't a complete implementation, but should give a sense of what's possible:

```

template <TemplateString S>
auto with_printf(S&& s) -> void {
    constexpr char const* fmt_string = [&]() consteval {
        std::string fmt = s.string(0);
        auto nsdms = nonstatic_data_members_of(remove_cvref(^^S),
            std::meta::access_context::current());

        for (size_t i = 0; i < s.num_interpolations(); ++i) {
            // in theory, this logic would be more interesting
            auto interp = s.interpolation(i);
            auto type = remove_cvref(type_of(nsdms[interp.index]));
            if (type == ^^int) {
                fmt += "%d";
            } else {
                if (interp.fmt == std::string_view("{:?}",)) {
                    fmt += "\"%s\"";
                } else {
                    fmt += "%s";
                }
            }

            fmt += s.string(i + 1);
        }

        return std::define_static_string(fmt);
    }();

    auto adjust = [<class T>(T const& arg){
        if constexpr (std::same_as<T, std::string>) {
            return arg.c_str();
        } else {
            return arg;
        }
    };

    auto& [...exprs] = s;
    constexpr auto [...Is] = std::make_index_sequence<s.num_interpolations()>();
    std::printf(fmt_string, adjust(exprs...[s.interpolation(Is).index])...);
}

```

The important thing is to expose all the relevant information to users to let them do whatever they want with it. Note that `highlighted_print` uses the `fmt`, `index`, and `count` fields of the interpolation, since it is formatting all of them, but not the expression field. Meanwhile, `into_json` uses only `expression` and `index` — it doesn't need any of the format specifier logic, since it isn't actually doing formatting. The `printf` example uses `string` to build up its own format specifier.

And of course, *all* of these operations can be performed on the same object:

```
auto verb() -> std::string { return "die"; }

auto main() -> int {
    std::string name = "Inigo Montoya";
    int width = 5;

    auto msg = t"Hello, my name is {name:~}. You killed my {relation}.
        Prepare to {verb():*^{width}}.\n";
    std::print(msg);
    highlighted_print(msg);
    std::println("{} ", into_json(msg));
    with_printf(msg);
}
```

will print (note that `highlighted_print` uses the format specifiers, so the name is quoted):

```
Hello, my name is "Inigo Montoya". You killed my father. Prepare to
    *die*.
Hello, my name is "Inigo Montoya". You killed my father. Prepare to
    *die*.
{"name":"Inigo Montoya","relation":"father","verb()":"die"}
Hello, my name is "Inigo Montoya". You killed my father. Prepare to die.
```

Having both `interp.index` and `interp.count` is a little clunky, especially since `interp.count` will almost always be `1`. But I think it's better to put the clunkiness there and maintain the trivial formatting implementations (where you can just unpack the template string object).

You can see this example on [compiler explorer](#). Note that the implementations there are slightly different, since Clang doesn't yet implement `constexpr` structured bindings and the implementations of `pack indexing` and `expansion statements` had a few bugs so I came up with workarounds.

§ 2.9 The TemplateString Concept

In these examples, I've been using this `TemplateString` concept to identify a template string object. The question is, what does that concept look like? This one I'm not sure about yet. It can't be a built-in, since users might need to create one of these objects. Consider a logger:

```
log::info(t"Got a trade for in {symbol}: {side} {qty} @ {price}");
```

The expressions here are all cheap to copy, but formatting is expensive — so I might want to serialize all of the data into a background thread to do my formatting there. I don't want to just copy the template string object, since it might have references. With reflection, I can create a new type that has only value members, and then keep the same `fmt`, `strings`, and `interpolations`:

```
template <TemplateString S, class F>
auto map(S s, F f) {
    struct Base;
    consteval {
        vector<meta::info> specs;
        for (meta::info m : nonstatic_data_members_of(^S)) {
            specs.push_back(data_member_spec({
                .type=invoke_result(^F, {type_of(m)}),
                .name=identifier_of(m),
            }));
        }
        define_aggregate(^Base, specs);
    }
    struct R : Base {
        static constexpr auto fmt = S::fmt;
        static constexpr auto string = S::string;
        static constexpr auto interpolation = S::interpolation;
        static constexpr auto num_interpolations = S::num_interpolations;
    };

    auto& [...pieces] = s;
    return R{{f(FWD(pieces))...}};
}
```

Which allows the implementation of all of the logging functions to map their provided template string object to decay or otherwise transform every member into something that won't dangle.

I'd want to make sure this `R` here is also considered a template string for all of these purposes. There are a few options here, not really sure which would be best:

- an attribute
- an annotation
- structural conformance: simply check for the presence of `fmt`, `string`, `num_interpolations`, and `interpolation`?

§ 2.10 Implementation Experience

I implemented this in Clang, on top of the p2996 reflection branch. Code can be found in my fork in the `template-strings` branch [here](#) (you can see the diff against p2996 [here](#)) I'm sure there are better ways to do some of what I did. It can also be used on [compiler explorer](#).

The only difference between this paper and the implementation is that instead of introducing the type `std::interpolation`, I just made `_Interpolation` implicitly defined at global scope for convenience.

This does raise the question of how `std::interpolation` should be defined. Should this facility really require a new header? Maybe it's a sufficiently trivial type (an aggregate with no member functions and just four data members, each of scalar type) that the compiler can just generate it? Maybe we don't care about additional headers because `import std`; anyway?

The same question goes for if we want to implement the [concept](#) by way of annotation. That annotation would be an empty type, could the compiler just create it?

§ 2.11 More Formal Lexing Specification

In Clang, preprocessing happens during lexing — and so the way I implemented it was that a template string lexes as a new kind of token — really a meta-token that itself contains a bunch of other tokens. That's fine as far as Clang goes (or maybe not, there might be a more preferred approach), but the standard defines the phases of translation (5.2 [\[lex.phases\]](#)) in a strict order. Specifically:

- Phase 3: Source file decomposed into preprocessing tokens
- Phase 4: Preprocessing directives and macro expansion occurs
- Phase 5: String-literal encoding and concatenation
- Phase 6: Tokens
- Phase 7: The rest of the owl

The wording needs to fit this specification. Which we can do by defining a new set of preprocessing tokens. For example, the template string

```
t"My name is {name():>{width}}.\n"
```

can lex into the tokens (indentation for clarity):

```
template-string-begin
  string-literal "My name is "

  template-string-interpolation-begin
    string-literal "{:>{}}"
    ,
    string-literal "name()"

    template-string-expression-begin
      identifier "name"
      (
      )
    template-string-expression-end

    template-string-expression-begin
      identifier "width"
    template-string-expression-end
  template-string-interpolation-end
```

```
string-literal ".\n"
template-string-end
```

That is, we have a bunch of meta-tokens that don't lead to any output and are just there to guide parsing. A template string literal lexes into this alternating sequence of *string-literals* and interpolations. An interpolation is bounded by *template-string-interpolation-begin* and *template-string-interpolation-end*, starts with two *string-literals* for the format string and expression (with a separator token just to avoid string concatenation from kicking in — I used `,` above but the actual separator doesn't matter), and then continues with at least one expression. An expression is bounded by *template-string-expression-begin* and *template-string-expression-end*.

To handle [trailing =](#), we could go one of two ways. Consider just `t"Got {x = }."`. That could be either:

Pre-emptively append	Delayed concat
<pre>template-string-begin string-literal "Got x = " template-string-interpolation- begin string-literal "{}" , string-literal "x" template-string-expression- begin identifier "name" template-string-expression- end template-string-interpolation- end string-literal "." template-string-end</pre>	<pre>template-string-begin string-literal "Got " // the initial string string-literal "x = " // expression with trailing = template-string-interpolation-begin string-literal "{}" , string-literal "x" template-string-expression-begin identifier "name" template-string-expression-end template-string-interpolation-end string-literal "." template-string-end</pre>

It works either way. With delayed concatenation, we rely on phase 5 to concatenate the string literals before parsing anyway.

We could conceivably also just introduce a single token and use braces for everything else. Though probably better to use something other than braces to make it easier to diagnose poorly formatted expressions? Whatever people prefer:

Several tokens for clarity	Just one token
<pre>template-string-begin string-literal "My name is " template-string-interpolation-begin string-literal "{:>{}" , </pre>	<pre>template-string { string-literal "My name is " { string-literal "{:>{}" , </pre>

Several tokens for clarity	Just one token
<i>string-literal</i> "name()"	<i>string-literal</i> "name()"
<i>template-string-expression-begin</i> <i>identifier</i> "name" () <i>template-string-expression-end</i>	{ <i>identifier</i> "name" () }
<i>template-string-expression-begin</i> <i>identifier</i> "width" <i>template-string-expression-end</i> <i>template-string-interpolation-end</i>	{ <i>identifier</i> "width" } }
<i>string-literal</i> ".\n" <i>template-string-end</i>	<i>string-literal</i> ".\n" }

§ 3 Alternate Approaches

This proposal is definitely not the only way to do string interpolation in C++. I've [already discussed](#) two previous proposals in this space and why I think what I'm proposing is a better design. But it's worth talking about other approaches as well.

§ 3.1 f-strings

This paper *only* proposes template strings, it does *not* propose a convenient shorthand for creating a `std::string`. If a `std::string` is desired, the user will have to write:

```
auto a = std::format("My name is {} and my age next year is {}", name,
    age+1); // status quo
auto b = std::format(t"My name is {name} and my age next year is
    {age+1}"); // proposed
auto c = f"My name is {name} and my age next year is {age+1}";
    // not proposed
```

The reasoning here is that moving from a to b is a significant gain in readability (as well as other functionality, as illustrated in previous examples), but the gain from b to c is simply saving a few characters. It's *nice*, but it comes at a heavy cost that I'm simply not sure is actually worth it. Is `std::format` specifically the most common formatting facility? I think small programs will have more calls to `std::print` or `std::println` while larger programs will have more calls to `std::format_to` as well as use logging. I just don't think the trade-off is there.

§ 3.2 Reduced Representation

What should this template string literal evaluate to?

```
t"New connection on {ip:#x}:{port}"
```

This paper proposes the one on the left, but we could just do the one on the right:

Proposed	Simpler
<pre>struct S { static constexpr auto fmt() -> char const*; static constexpr auto num_interpolations() - > size_t; static constexpr auto string(size_t n) -> char const*; static constexpr auto interpolation(size_t n) -> interpolation; u32 _0; u16 _1; };</pre>	<pre>struct S { static constexpr auto fmt() - > char const*; u32 _0; u16 _1; };</pre>

But the main motivation (and likely the most common use-case) is some version of formatting, for which all we need is `S::fmt()`. We wouldn't need `S::interpolation(n)` or `S::string(n)`. Should we still generate the extra functions?

I would argue that we should. The implementation has to do all the work to get those pieces anyway (with the exception of the index and count members for each interpolation, which really isn't much work), so it's not like we're saving much in the way of computation by stripping the interface. The simpler interface is only simple in that it reduces the available functionality. Doesn't seem like a good idea.

§ 3.3 Redundant Information

Continuing with the previous example, what if instead of removing `S::string` and `S::interpolation`, we instead removed `S::fmt`?

Proposed	Minimal
<pre> struct S { static constexpr auto fmt() -> char const*; static constexpr auto num_interpolations() -> size_t; static constexpr auto string(size_t n) -> char const*; static constexpr auto interpolation(size_t n) -> interpolation; u32 _0; u16 _1; }; </pre>	<pre> struct S { static constexpr auto num_interpolations() -> size_t; static constexpr auto string(size_t n) -> char const*; static constexpr auto interpolation(size_t n) -> interpolation; u32 _0; u16 _1; }; </pre>

Note that `S::fmt()` is exactly the result of concatenating `S::string(0)`, `S::interpolation(0).fmt`, `S::string(1)`, `S::interpolation(1).fmt`, and `S::string(2)`. This is true by construction for all template strings, and is precisely how it is implemented as well. Given that `S::string(n)` and `S::interpolation(n)` will both exist (as being more fundamental), do we need to also provide `S::fmt()` — which can simply be derived from both arrays?

One advantage of removing `S::fmt()` is to avoid having that string spill into the binary even if unused, if the implementation simply fails to detect its lack of use. However, I think we should. While formatting will not be the *only* usage of these objects, it is both the main motivating and primary one, so it will both be more convenient for users and more efficient if the compiler simply does that little bit of extra work to produce the full format string as well.

And regardless, the opposite problem would still exist anyway — if only `S::fmt()` were used, there is the potential that the string literals in `S::string(n)` and `S::interpolation(n)` spill into the binary unnecessary as well.

§ 3.4 Static Data Members or Static Member Functions

The proposal right now has 4 static member functions. But a simple alternative would be to instead provide three, with `strings()` and `interpolations()` returning `std::span` instead of having a `string(n)` and `interpolation(n)` and `num_interpolations()`:

```

struct S {
    static constexpr auto fmt() -> char const*;
    static constexpr auto strings() -> std::span<char const* const>;
    static constexpr auto interpolations() -> std::span<std::interpolation const>;
};

```

The advantage of approach is that it allows directly looping over the interpolations via:

```
template for (constexpr auto interp : s.interpolations()) {
    // ...
}
```

Which instead some of the [examples](#) work around via:

```
template for (constexpr auto I : std::views::indices(s.num_interpolations())) {
    constexpr auto interp = s.interpolation(I);
}
```

The disadvantage is that it brings in the `std::span` dependency. But if we're declaring `std::interpolation` anyway, that's probably not that big a deal.

A very different shape might instead be to have static *data members* instead of static functions, where:

```
struct S {
    static constexpr char const* fmt = /* ... */;
    static constexpr char const* strings[] = /* ... */;
    static constexpr std::interpolation interpolations[] = /* ... */;
};
```

This would avoid the span dependency and avoid a bunch of parentheses that you would have to write, as compared to the other function version. However, it has two problems. First, a template string can have no interpolations, and we still don't have zero-sized arrays in C++. We ran into this problem with specifying `std::meta::reflect_constant_array`, and it's very annoying that it doesn't just work (even as gcc and clang happily support them with expected semantics with no warnings). Neither the version where `interpolations()` returns a `std::span` nor the version where we have `interpolation(n)` which returns the *n*th interpolation have this problem.

The second problem is that these types may have to be local types in some contexts, and local types cannot have static data members in C++. I do not know why local types have this restriction, given that local types are allowed to have static member functions and those member functions are allowed to have static local variables. But the restriction does currently exist.

§ 3.5 Wait for Reflection

The evergreen question with proposals like this is to wonder if we should wait for reflection. There even is a proposal, [\[P3294R2\] \(Code Injection with Token Sequences\)](#), that has walks through how a future macro could solve this problem. Concretely, we would need:

- macros that can inject token sequences,
- the ability to invoke such a macro with a string literal (or `string_view`), and
- the ability to turn a `string_view` into a token sequence

Given those pieces, the implementation of something like Rust's `format_args!` is basically the same as how you would implement it in a compiler. And the benefit of being able to do this in a library is pretty clear: it is much easier to experiment with different functionality. Plus, this would just be a very small taste of what code injections could do.

So should we wait? It seems incredibly unlikely that we will land something as expansive as token sequence injection in C++29 (if ever?), and a dedicated language feature for template string objects is pretty small and self-contained.

§ 4 References

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